



Exploring the Dynamics of Pipelining

Pipelining is a technique used in the design of computers and other digital electronic devices to increase the instruction throughput. In this article, we will look at the concept behind pipelining.

Pipelining is an implementation technique where multiple instructions are overlapped in execution. The computer pipeline is divided into stages that operate in parallel. They are connected, one to the next, to form a pipe. Instructions enter at one end, progress through the various stages, and exit at the other end.

As mentioned earlier, the pipelining concept was devised to increase instruction throughput

(the number of instructions that can be executed in a unit of time). The idea behind this is to split the processing of a computer instruction into a series of independent steps, with storage at the end of each step. This allows the computer's control circuitry to issue instructions at the processing rate of the slowest step, which is much faster than the time needed to perform all steps at once. The term 'pipeline' refers to the fact that each step is carrying data at the same time (like how a pipe carries water), and each